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Contact: janhavi.mh@gmail.com
Company: CarWale
Company status: Dream
CTC: 9 Lacs
Job location: Vashi, Navi Mumbai
Job Offered: Yes

Profile	Academic Eligibility	Disciplines	Branches	Compensation
Associate Software Development Engineer	7 & above CGPA or equivalent. No Active Backlog	MCA / BE/B TECH/ME /MTECH	Computer Science ,Information Technology, Electrical, Electronics	CTC – INR 9,00,000/- per annum Stipend fee - 25000/-
Front End Developer	7 & above CGPA or equivalent. No Active Backlog	MCA / BE/B TECH/ME /MTECH	Computer Science ,Information Technology, Electrical, Electronics	CTC – INR 9,00,000/- per annum Stipend fee - 25000
Associate Software Development in Test	7 & above CGPA or equivalent. No Active Backlog	MCA / BE/B TECH/ME /MTECH	Computer Science ,Information Technology, Electrical, Electronics	CTC – INR 9,00,000/- per annum Stipend - 25000

Test:

Date: 6th August, 2022

Time limit: 2 hours

The test was conducted on **Hackerrank** at **3 pm**. There were 4 questions (1 easy, 2 medium and 1 hard). I solved 2 questions with all test cases passed, and one where half of test cases passed.

Q1) Data for n days is given. 'k' updates are performed on that data where each update is in the form [l, r]. This indicates that data starting at index 'l' and ending at index 'r' is negated. Given the initial data and 'k', find final data after updates.

Q2) For an array 'arr' of 'n' integers following operation can be performed for any number of times:

- 'k' ($1 \leq k \leq m$) elements can be removed from the array if $2^{arr[1]} + 2^{arr[2]} + \dots + 2^{arr[k]} = 2^p$ for some non negative integer p.

Given an array 'arr' of 'n' integers, find min number of operations to remove all elements from array.

Q3) Choose k non-empty subarrays from array 'arr' of 'n' integers. sub_sum[i] is the sum of elements in ith subarray. The beauty of 'arr' is $1 * \text{sub_sum}[1] + 2 * \text{sub_sum}[2] + \dots + k * \text{sub_sum}[k]$. For given 'arr' and 'k', find the max possible beauty of the array.

45 students were shortlisted, and the result was declared on the same day.

Interview:

Date: 8th August, 2022

The interviews started in the order of the students that were shortlisted. All the rounds were going on simultaneously. The interviews started at 9 am, and the results were announced at around 8 pm. All the interviewers were helpful and cooperative.

Technical Round 1 (DSA & Coding):

This round was purely DSA and coding based. It started with the interviewer asking me to introduce myself.

Q - You are given an array of integers ranging from 1 to N. There are no duplicates in the list. One of the integers is missing in the array. The task is to find the missing number in the series. Example: Given array [1, 2, 4, 5], find the missing number.

Answer: 3

Logic:

I first answered the brute force method (basically comparing a given array with an array of numbers 1 to N and finding the missing number) then he asked me for a more efficient algorithm for the same and I suggested the method using the sum of numbers from 1 to N can be calculated by using the formula $N * (N + 1) / 2$. Now find the summation of all elements in the array and subtract it from the summation of first N natural numbers, the value obtained will be the value of the missing element.

The interviewer then asked me to solve the question using the binary search algorithm. I then implemented the binary search algorithm, and split the array into two parts and so on. He then asked me to present my screen and write the code using the given logic. He then asked me some questions in the code.

Tips:

- Do dynamic programming and DSA.
- It's okay if you don't know the code, just tell the algorithm.

Technical Round 2 (System Architecture):

The interviewer was the system architect of CarWale. He started by asking me to introduce myself. He then asked me how my previous interview was. He then read my resume and asked me about my projects. As my profile was more on the frontend side, he asked me a bit about the

process of designing a website. He then went over to the more technical side, and asked me how a website works and to explain caching. He then asked me to design the system of databases for the CarWale website. He asked questions like: how will you handle load on the server and what will you do if you have multiple requests creating load on the server? Lastly he asked me if I had any questions, so I asked him for feedback on my interview, and how I can improve myself.

HR Round:

The interview was taken by the CTO of CarWale.

- Introduce yourself
- Summarize what all life lessons learnt in your school
- Who was your best friend in school and how would she describe you?
- Summarize what all life lessons learnt in 11 and 12th
- What are your 3 strengths and weaknesses?
- Summarize what all life lessons learnt in college

Lastly, he asked if I had any questions, I asked him what tips he would give to someone who is just starting out as a developer.

Total 6 students were selected. All the best, you guys!